

FLEET TRACKING DASHBOARD — PROJECT SPECIFICATION (MAPUP ASSESSMENT)

🎯 OBJECTIVE

BUILD A REAL-TIME FLEET TRACKING DASHBOARD THAT VISUALIZES VEHICLE MOVEMENT AND OPERATIONAL METRICS USING SIMULATED STREAMING TRIP DATA.

THE SYSTEM SHOULD BEHAVE LIKE A LIVE FLEET MONITORING PLATFORM, PROCESSING EVENT STREAMS AND UPDATING UI IN REAL TIME.

🚚 SYSTEM OVERVIEW

THE APPLICATION WILL TRACK 5 SIMULTANEOUS VEHICLE TRIPS, EACH WITH INDEPENDENT EVENT STREAMS:

- CROSS-COUNTRY LONG HAUL (10,000+ EVENTS)
- URBAN DENSE DELIVERY (500+ EVENTS)
- MOUNTAIN ROUTE CANCELLED (100+ EVENTS)
- SOUTHERN TECHNICAL ISSUES (1,000+ EVENTS)
- REGIONAL LOGISTICS (2,000+ EVENTS)

EACH TRIP EMITS TIMESTAMPED EVENTS THAT MUST BE PROCESSED IN CHRONOLOGICAL ORDER TO SIMULATE REAL-TIME MOVEMENT.

⚙️ CORE REQUIREMENTS

1. REAL-TIME EVENT PROCESSING

- PROCESS EVENTS IN CHRONOLOGICAL ORDER
 - SIMULATE LIVE STREAMING USING:
 - `setInterval()` OR
 - EVENT QUEUE PROCESSING OR
 - MOCK WEBSOCKET SERVER
 - MAINTAIN A GLOBAL SIMULATION CLOCK
 - SUPPORT REPLAY / SPEED CONTROL:
 - 1x, 5x, 10x SPEED
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2. STATE MANAGEMENT

TRACK AND UPDATE:

PER VEHICLE / TRIP

- CURRENT LOCATION (LAT/LNG)

- SPEED
- STATUS (MOVING / STOPPED / IDLE / ERROR / COMPLETED / CANCELLED)
- DISTANCE TRAVELED
- FUEL LEVEL (IF APPLICABLE)
- LAST EVENT TIMESTAMP

FLEET LEVEL

- TOTAL ACTIVE TRIPS
- COMPLETED TRIPS
- CANCELLED TRIPS
- TRIPS > 50% PROGRESS
- TRIPS > 80% PROGRESS
- AVERAGE SPEED ACROSS FLEET

3. DASHBOARD UI (FRONTEND)

BUILD A RESPONSIVE DASHBOARD WITH:

📍 MAIN VIEWS

1. FLEET OVERVIEW PANEL
 - TOTAL TRIPS
 - ACTIVE / COMPLETED / CANCELLED
 - PROGRESS DISTRIBUTION
2. TRIP LIST PANEL
 - ALL 5 TRIPS WITH STATUS INDICATORS
 - PROGRESS BARS
 - QUICK STATS PER TRIP
3. TRIP DETAIL VIEW
 - MAP VISUALIZATION (LIVE ROUTE TRACKING)
 - EVENT TIMELINE
 - CURRENT VEHICLE STATUS
 - ALERTS / ISSUES
4. LIVE EVENT FEED
 - STREAMING EVENT LOG

- COLOR-CODED EVENT TYPES (INFO, WARNING, ERROR)
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4. MAP VISUALIZATION

- SHOW VEHICLE MOVEMENT ON MAP (GOOGLE MAPS / MAPBOX / LEAFLET)
 - DRAW POLYLINE FOR ROUTE HISTORY
 - ANIMATE VEHICLE MARKER BASED ON EVENTS
 - UPDATE POSITION IN REAL-TIME
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5. EVENT SYSTEM

YOU WILL PROCESS 27 EVENT TYPES, SUCH AS:

- START_TRIP
- LOCATION_UPDATE
- SPEED_CHANGE
- FUEL_UPDATE
- ENGINE_ERROR
- ROUTE_DEVIATION
- TRIP_COMPLETED
- TRIP_CANCELLED

EACH EVENT CONTAINS:

- TIMESTAMP
 - TRIPID
 - EVENTTYPE
 - PAYLOAD (DYNAMIC JSON)
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📄 ARCHITECTURE REQUIREMENTS

FRONTEND (RECOMMENDED)

- REACT / NEXT.JS (PREFERRED)
- TYPESCRIPT (RECOMMENDED)
- STATE MANAGEMENT:
 - REDUX OR ZUSTAND OR CONTEXT API
- MAP LIBRARY:

- LEAFLET / MAPBOX / GOOGLE MAPS

BACKEND (OPTIONAL BUT RECOMMENDED)

YOU MAY INCLUDE:

- NODE.JS + EXPRESS SERVER
- STREAMING ENDPOINT:
 - /API/EVENTS/STREAM
- WEBSOCKET SERVER FOR REAL-TIME PUSH

OR SIMULATE FULLY ON FRONTEND USING JSON FILES.

⚡ PERFORMANCE REQUIREMENTS

- MUST HANDLE 10,000+ EVENTS EFFICIENTLY
- USE:
 - MEMOIZATION
 - BATCHING UPDATES
 - THROTTLED UI RENDERING
- AVOID UNNECESSARY RE-RENDERS

KEY INSIGHTS TO DISPLAY

DASHBOARD SHOULD HIGHLIGHT:

- TRIP COMPLETION %
- AVERAGE FLEET SPEED
- DELAYED / FAILED TRIPS
- FUEL CONSUMPTION TRENDS (IF AVAILABLE)
- ROUTE DEVIATIONS
- ERROR FREQUENCY PER TRIP

SIMULATION CONTROLS

ADD UI CONTROLS:

-  START / PAUSE SIMULATION
-  SPEED CONTROL (1x, 5x, 10x)

-  RESET SIMULATION
 -  REPLAY FROM START
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UX REQUIREMENTS

- CLEAN, ENTERPRISE-STYLE DASHBOARD UI
 - DARK MODE PREFERRED
 - RESPONSIVE (DESKTOP-FIRST, TABLET SUPPORTED)
 - REAL-TIME SMOOTH ANIMATIONS
 - COLOR-CODED STATUS SYSTEM:
 - GREEN → NORMAL
 - YELLOW → WARNING
 - RED → CRITICAL/ERROR
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DEPLOYMENT

DEPLOY TO:

- VERCEL / NETLIFY (FRONTEND)
- RENDER / RAILWAY (BACKEND IF USED)

MUST INCLUDE:

- PUBLIC URL
 - WORKING LIVE SIMULATION
 - NO BROKEN ASSETS
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PROJECT STRUCTURE (SUGGESTED)

```
/FLEET-DASHBOARD
/FRONTEND
/COMPONENTS
/PAGES
/HOOKS
/STORE
/UTILS
/MAPS
/BACKEND (OPTIONAL)
/ROUTES
/SERVICES
/WS
```

/DATA
TRIPS.JSON
EVENTS/
README.MD

📖 COPILOT INSTRUCTION (IMPORTANT)

WHEN GENERATING CODE, ENSURE:

- SYSTEM BEHAVES LIKE A REAL-TIME STREAMING FLEET TRACKER
 - EVENT PROCESSING IS TIME-BASED, NOT INSTANT RENDER
 - UI UPDATES ARE INCREMENTAL, NOT FULL RELOAD
 - ARCHITECTURE SUPPORTS SCALABILITY TO LARGE EVENT DATASETS
 - CODE IS MODULAR AND PRODUCTION-GRADE
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🚩 FINAL OUTPUT

A FULLY WORKING:

🖥️ REAL-TIME FLEET TRACKING DASHBOARD WITH SIMULATED STREAMING DATA, MAP VISUALIZATION, AND FLEET ANALYTICS.